A Comparative Study of GAN-Based Image-to-Image Translation Methods for AFM-to-O2A Microscopy Images

Tahir Kurtar - Assist. Prof. Dr. Başak Esin Köktürk Güzel

Izmir Democracy University – Electrical and Electronics Engineering

Abstract

This study evaluates GAN-based image-to-image translation methods for converting AFM images into O_2A amplitude maps. A supervised Pix2Pix model is used as the baseline and compared with CycleGAN, ESRGAN-based pipelines, GauGAN, and a vanilla GAN. Results show that GauGAN achieves the best quantitative performance (L_1, SSIM, PSNR), while other models offer trade-offs between visual quality and numerical accuracy, demonstrating the feasibility of GAN-based AFM-to-O_2A synthesis.

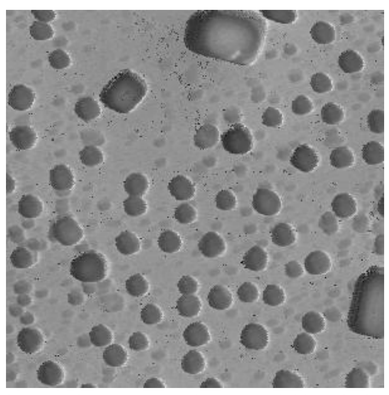
Introduction

Image-to-image translation using Generative Adversarial Networks (GANs) has enabled effective cross-modality mapping while preserving structural information. In nanoscale microscopy, Atomic Force Microscopy (AFM) provides high-resolution topography, whereas Second Harmonic s-SNOM amplitude (O_2A) images reveal complementary optical contrast but require costly and less accessible instrumentation. This study investigates whether O_2A amplitude maps can be synthetically generated from standard AFM images using deep learning. A supervised Pix2Pix model is established as a baseline and systematically compared with CycleGAN, ESRGAN-enhanced pipelines, GauGAN, and a vanilla GAN to evaluate their suitability for AFM-to-O_2A translation. Model performance is assessed through quantitative metrics (L_1 loss, SSIM, PSNR) and qualitative visual analysis, aiming to identify the strengths and limitations of different GAN architectures for nanoscale modality translation and to reduce reliance on specialized optical imaging systems.

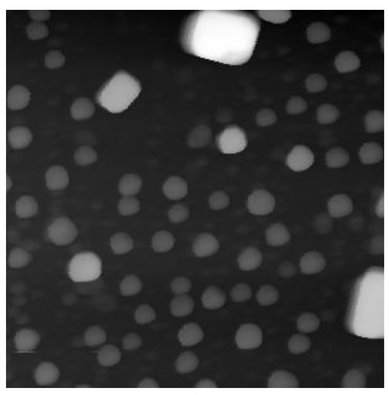
Materials and Methods

Dataset

The dataset consists of paired Atomic Force Microscopy (AFM) images and corresponding Second Harmonic scattering-type Scanning Near-Field Optical Microscopy (O_2A) amplitude maps acquired from the same nanoscale regions. AFM images provide surface topography, while O_2A maps capture optical amplitude contrast. All image pairs are spatially aligned and normalized to the range [−1, 1]. The dataset is split into training, validation, and test sets, with the test set held out for final evaluation.



Second Harmonic scattering-type Scanning Near-Field Optical Microscopy (O_2A)



Atomic Force Microscopy (AFM)

Model Architectures

Multiple Generative Adversarial Network (GAN) architectures are implemented and compared for AFM-to-O_2A image translation:

- **Pix2Pix (Supervised baseline):** U-Net generator with PatchGAN discriminator trained on paired data using adversarial and L_1 loss.
- **CycleGAN:** Unpaired translation using cycle-consistency constraints.
- **ESRGAN-enhanced pipeline:** Super-resolution refinement applied to generated O_2A images.
- **GauGAN:** Spatially-aware synthesis guided by structural information.
- **Vanilla GAN:** Simple adversarial baseline without reconstruction loss.

Training Setup

All models are trained using the Adam optimizer with fixed hyperparameters for fair comparison. Training is performed for a fixed number of epochs with early stopping based on validation performance. Experiments are conducted using the PyTorch framework on GPU-enabled hardware.

Evaluation Metrics

Model performance is evaluated on the held-out test set using:

- **L_1 error** for pixel-wise accuracy,
- **Structural Similarity Index (SSIM)** for perceptual similarity,
- **Peak Signal-to-Noise Ratio (PSNR)** for signal fidelity.

Qualitative visual comparisons are also performed to assess structural consistency and texture realism.

AFM → GAN Model → Synthetic O_2A → Metrics (L1 / SSIM / PSNR)

Results

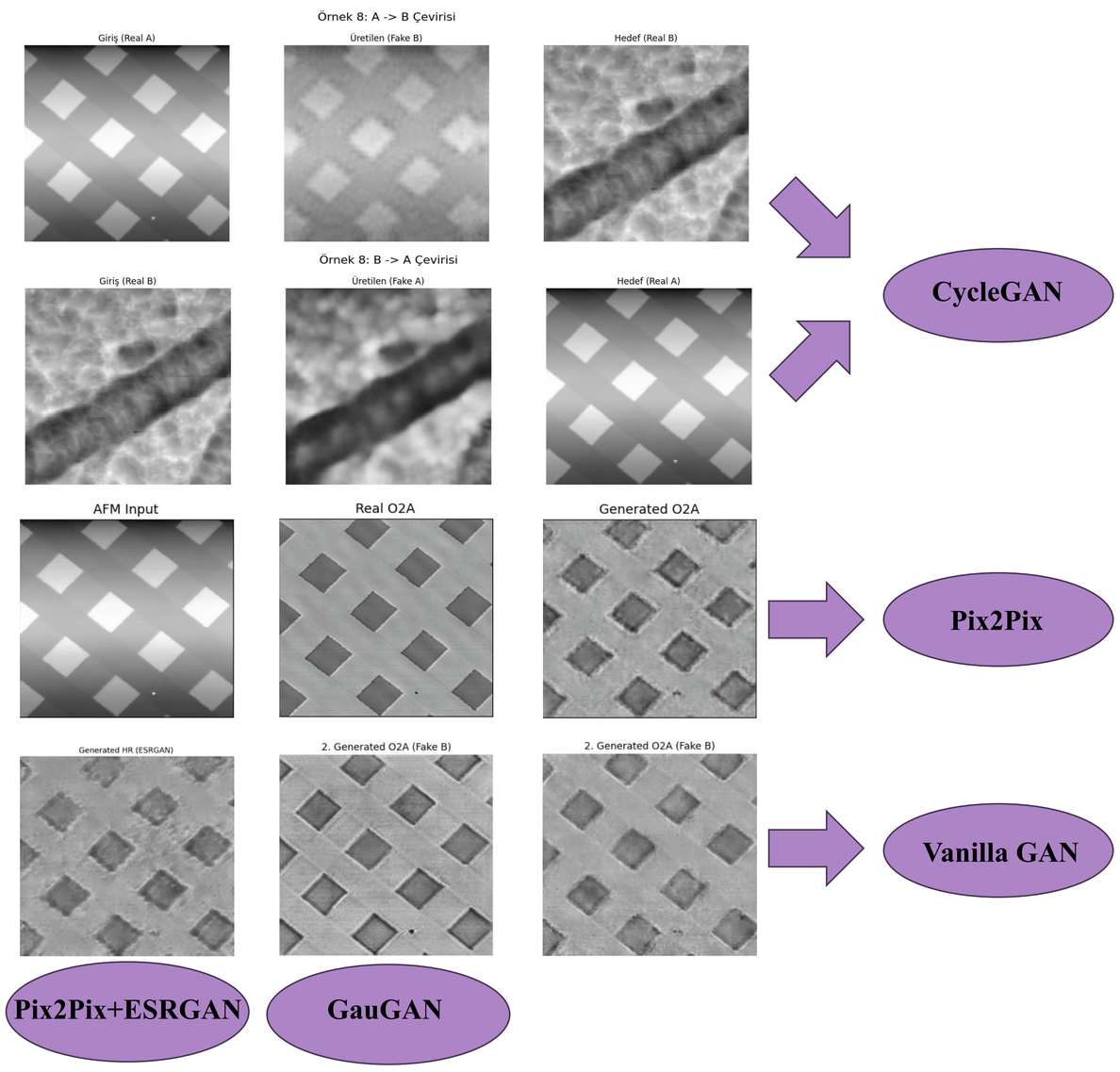
Quantitative Results

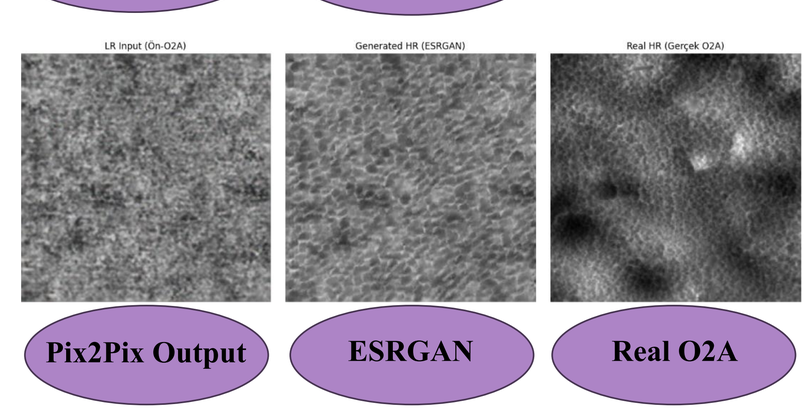
GauGAN achieves the best overall performance, yielding the lowest L_1 loss, highest SSIM and PSNR, and the most visually coherent results. **Pix2Pix** demonstrates stable and competitive quantitative performance, confirming the effectiveness of paired supervision. **ESRGAN** mainly enhances visual sharpness without consistent improvements in numerical accuracy. **CycleGAN** exhibits significantly lower quantitative performance due to its unpaired training strategy. The **Vanilla GAN** shows the weakest results in both quantitative metrics.

	Pix2Pix	Pix2Pix + ESRGAN	GauGAN	Vanilla GAN	Cycle GAN A--->B	Cycle GAN B--->A
L1 Loss	0.2084	0.2062	0.202	0.2054	1.8326	1.8326
SSIM	0.4422	0.4419	0.457	0.4255	0.2421	0.2216
PSNR	19.48 dB	19.39 dB	19.49 dB	19.22 dB	12.86 dB	11.04 dB

Qualitative Results

Visual comparisons indicate that **GauGAN** produces the most perceptually realistic O_2A maps, while **Pix2Pix** and the **Vanilla GAN** exhibit comparable visual quality, with the **Vanilla GAN** occasionally yielding sharper results. **CycleGAN** captures coarse structures but introduces smoothing artifacts, and **ESRGAN** enhances sharpness without improving structural fidelity.





ESRGAN enhances sharpness without improving structural fidelity

Conclusions

Quantitative evaluation shows that **GauGAN achieves the best overall performance**, yielding the lowest L_1 loss and the highest SSIM and PSNR values. **Pix2Pix** remains a strong and stable supervised baseline, while **ESRGAN** mainly improves visual smoothness without enhancing feature learning. Qualitative analysis reveals that **GauGAN produces the sharpest and most visually coherent O_2A images**, whereas **Pix2Pix** and the **Vanilla GAN** exhibit comparable visual quality, with the **Vanilla GAN** occasionally producing sharper results. **CycleGAN** exhibits limited performance due to its unpaired training structure. Overall, GAN-based methods demonstrate the feasibility of AFM-to-O_2A synthesis without specialized s-SNOM instrumentation.